

Total No. of Questions : 10]

SEAT No. :

P3524

[4858]-1006

[Total No. of Pages : 3

T.E. (Civil)

ADVANCED SURVEYING

(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.No. 1 or Q. No. 2, Q.No. 3 or Q. No. 4, Q.No. 5 or Q. No. 6, Q.No. 7 or Q. No. 8, Q.No. 9 or Q. No. 10.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) State the objects of Geodetic Surveying and explain Secondary Triangulation? **[5]**

b) Describe three important segments of GPS system with illustration. **[5]**

OR

Q2) a) Differentiate between plane surveying and Geodetic surveying. **[5]**

b) What are various potential errors sources that affect GPS signal or result?**[5]**

Q3) a) Explain the analytical method of solving three point problem. **[5]**

b) Write a note on correction for curvature and refraction. **[5]**

OR

Q4) a) Explain and calculate data for intersecting circle method of graphical Solution. **[6]**

b) Describe in brief location survey of a long bridge. **[4]**

Q5) a) What do you mean by a spherical triangle and how do you find out the length of sides of a spherical triangle? **[5]**

b) Define Geodetic quadrilateral and describe methods of its adjustment.**[5]**

P.T.O.

- c) Find the most probable values of the angles A,B and C of a triangle ABC from the following observations (Use method of differences). [8]

Angle	Weight
Angle A = $65^{\circ} 15' 30''$	3
Angle B = $51^{\circ} 11' 25''$	2
Angle C = $63^{\circ} 32' 34''$	4

OR

- Q6)** a) Define following terms [5]

- i) True value,
- ii) Most probable value,
- iii) Conditioned Quantity
- iv) mistakes,
- v) Weight of an observation

- b) Explain clearly what is meant by side equation. How would you adjust a geodetic quadrilateral (without central station). [5]

- c) Four angles are measured at a station closing the horizon. The values of the angles are : [8]

A = $83^{\circ} 42' 28.75''$	weight 3
B = $102^{\circ} 51' 43.26''$	weight 2
C = $94^{\circ} 38' 27.22''$	weight 4
D = $79^{\circ} 23' 23.27''$	weight 2

Give the corrected values of the angles. (use Normal equation)

- Q7)** a) Define and explain the following terms with neat sketch : [5]

- i) Exposure station
- ii) swing
- iii) Azimuth
- iv) principal line
- v) Tilt

- b) What are the different types of aerial photographs? Discuss any one briefly giving their use. [4]

- c) A section line AB appears to be 10.16 cm on photograph for which the focal length is 16 cm. The corresponding line measures 2.54 cm on a map which is to a scale 1/15000. The terrain has an average elevation of 200m above mean sea level. Calculate flying height of the aircraft, above mean sea level, when the photograph was taken. [7]

OR

- Q8)** a) Explain Relief displacement and write the conclusions that can be drawn from expression of relief displacement. [5]
- b) What are the different stereo viewing techniques in digital photogrammetry? [4]
- c) A line AB 2000m long, lying at an elevation of 500m measures 8.65 cm on a vertical photograph for which focal length is 20 cm. Determine the scale of the photograph in an area the average elevation of which is about 800 m. [7]

- Q9)** a) What are the functionalities of GIS? [5]
- b) Give the application of remote sensing with respect to natural hazards and that of archaeology. [5]
- c) Explain the advantages and disadvantages of the raster and vector data models. [6]

OR

- Q10)** a) What are the components of a GIS? [5]
- b) Define remote sensing and enlist the advantages and limitations of remote sensing. [5]
- c) Explain the applications of GIS in Visibility analysis and slope analysis. [6]

